Gravitas

User Manual

An interactive astrophysics sandbox and astronomy teaching tool that runs entirely in the browser

https://gravitas-sim.online

59 scenarios • 22 guided investigations • 618 steps

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Contents

1 What Gravitas is

Gravitas is a gravity simulation you can teach with. It runs in a browser, needs no account and no installation, and is served as static files: nothing you do in it reaches a server, and nothing about you is recorded by it.

It is two things at once. It is a *sandbox* — place stars, planets and black holes, give them velocities, and watch what Newtonian gravity does with them — and it is a set of *guided investigations* that use that sandbox to take a student from a prediction, through a measurement, to a number they can defend.

1.1 What it models, and what it only draws

This distinction matters more than any other in this manual, and the application states it in public at https://gravitas-sim.online/model/.

Calculated
: Newtonian gravitation between every pair of bodies, in two dimensions, integrated with a scheme you choose. Orbits, energies, angular momenta, transit depths, radial-velocity curves, astrometric wobble, tidal differentials and rotation curves all come out of that integration.

Approximated
: Mergers are perfectly inelastic. The gravitational-wave inspiral is a phenomenological energy loss, not a solution of general relativity. The dark-matter halo is an added force term with an adjustable flat rotation speed.

Drawn only
: Accretion-disc glow, relativistic jets, gravitational-lensing distortion and the bloom around bright objects are pictures. They influence nothing.

Every claim the simulation makes about a measurable quantity is checked by a validation suite of 243 deterministic tests, each with a stated tolerance and a written reason for that tolerance. The results, and the list of things that are deliberately *not* validated, are published with the project.

2 Getting started

Open https://gravitas-sim.online. The simulation starts immediately; a first visit also shows a short front door offering three ways in, which you can dismiss and never see again.

2.1 The screen

The simulation
: fills the window. Scroll to zoom, drag to pan, click a body to select it.

The readout, top left
: carries the live numbers: elapsed simulated time, the zoom, the simulation speed, how many bodies of each kind are present, and — if you ask for them in Settings — the integrator in use and how far energy and angular momentum have drifted since the run began.

The control rail, right
: is an accordion: Scenario, State, Tools, View, Capture and Learn. One section is open at a time.

The transport bar, bottom
: pauses, steps and scrubs. The simulation records its own history, so the scrubber rewinds through what has already happened and holds on the frame you land on until you return to live.

The scale bar, bottom left
is always on the canvas, not in the page. It is drawn onto the simulation itself so that a screenshot carries its own scale.

2.2 Keyboard

Key	What it does
`Space`	Pause and resume
`R`	Recentre the camera and return to 1× zoom
`P`	Take a screenshot
`E`	Export the recorded run as CSV
`?`	The full shortcut list
`Esc`	Leave lecture mode, or close the open dialog

The complete list is in the Shortcuts dialog, which is the authoritative copy; this table is the handful you will use in a first session.

3 The instruments

Gravitas is a measuring instrument before it is a display. Everything in this section is in the Tools section of the rail.

3.1 Ruler, protractor, stopwatch

The **ruler** is dragged across the scene and reads a distance in AU and in kilometres. The **protractor** has a vertex and two arms and reads the angle between them. Both store their handles in *world* coordinates, so panning or zooming does not change what they say — a ruler pinned to two bodies keeps measuring those two bodies.

The **stopwatch** runs on simulated time, so pausing pauses it and changing the simulation speed does not change what it measures. With a body selected it can be latched to that body's periapsis passages: each lap is then one orbit, timed from closest approach rather than by your reaction time, and the panel reports the mean of the laps.

3.2 The object inspector

Select a body and the inspector gives its mass, radius, speed, the orbital elements of its path about whatever it is actually orbiting — semi-major axis, eccentricity, period, periapsis and apoapsis — and its kinetic, potential and total energy.

3.3 Why the bodies are not to scale

The bodies are drawn on a compressed scale, and the canvas says so: a line beside the scale bar reads *Body sizes enlarged for visibility.* It is drawn on the canvas rather than in the interface, so it is in every screenshot and every recording you make.

The real ratios cannot be drawn. The Sun's radius is about 109 times the Earth's; at a zoom where the Sun is a visible disc, an Earth drawn to scale beside it is a fraction of a pixel. Gravitas compresses the hierarchy instead, by one documented policy applied to every scenario: a Sun-like star is drawn about eight times the radius of an Earth-like planet and four times a Jupiter-like giant, against real ratios of 109 and 9.7. Gas giants are visibly larger than rocky planets, white dwarfs smaller than ordinary stars, and neutron stars, comets, asteroids and distant planets sit at a minimum marker size because at their true relative sizes they would not be visible at all.

Read sizes from the inspector and from the investigation measurements, never off the drawing. The number in the inspector is the physical radius; what is on the canvas is an illustration of it. The scale *bar* is a different matter and is an honest statement about distance in the current view — though some scenarios are themselves scale models or teaching constructions, so not every drawn distance is physical either. The model page at https://gravitas-sim.online/model/ sets all of this out under *Displayed sizes*, including the four different radii a body has and which of them each part of the simulation uses.

A body is no harder to select for being drawn small: the click target is computed from the body's own size and a minimum in screen pixels, and has nothing to do with how large the body appears.

About a quarter of procedurally generated gas giants are drawn with a ring system. That fraction is a choice about visual variety, not a claim about how common rings are among giant exoplanets, which nobody knows. Which giants get them, and what the rings look like, is fixed by each body's own seed, so it is the same after a reset, a reload, a shared link or a screenshot; scenarios may force rings on or off, and Saturn is ringed in the Solar System. Rings are drawn only: they have no mass and take no part in gravity, collisions, transit depth or any measurement.

3.4 Vectors and the potential well

For the selected body, Gravitas can draw its velocity, its total acceleration, and one arrow per gravitational source acting on it, each in a colour that cannot be confused with the others. The arrows are the acceleration the integrator actually used, not a second calculation that might disagree with it.

They exist for one misconception in particular: on an eccentric orbit velocity and force are perpendicular at periapsis and apoapsis and never come within forty degrees of parallel anywhere. An optional potential-well underlay shades the whole scene by gravitational potential.

3.5 Reference frames

The Frame selector re-expresses every position and every trail as a chosen body would see them. This is not the same as following a body with the camera: the recorded paths are redrawn. Put the Solar System into Earth's frame and Mars traces the retrograde loop that the geocentric model was built to explain.

4 The observing panels

Three panels turn the simulation into an observation, and all three are driven by the same viewing geometry. Drag the observer handle on the canvas, or use the angle controls, and every open panel re-measures from the new direction.

Transit light curve
Brightness against time for the starlight reaching the observer. A planet crossing its star carves a dip; the depth gives the planet's radius, and limb darkening is included, so the dip is not flat. The depth is $(R_p/R_s)^2$ computed from the *physical* radii — the ones the inspector reports, derived from mass — and not from the enlarged markers on the canvas. A planet drawn an eighth of its star's width does not produce a dip of one sixty-fourth; measure the depth from the curve and check it against the inspector's radii, never against the picture.

Radial velocity
The star's motion toward and away from the observer, as a spectrograph measures it. The wob-

ble gives a planet mass — times the sine of the inclination, which is where the $\sin i$ degeneracy stops being a footnote and becomes something a student can see by tilting the system.

Astrometry
The star's position on the sky over time, and the astrometric signature at an assumed distance.

Rotation curve
Orbital speed against distance from the centre, one point per body, drawn against the curve the visible mass alone would produce.

Open panels stack in the bottom-left corner rather than covering each other. The spacetime view — an optional 3-D rendering of the scene as a rubber sheet dipping around each mass — joins the same stack, and can be dragged out of it.

5 Choosing the physics

5.1 Integrators

Three schemes, switchable while the simulation runs, with live energy and angular-momentum drift beside them:

Symplectic Euler
The default, and what every scenario in the catalog is tuned against. First order, and its energy error is bounded rather than growing without limit.

Velocity Verlet
Second order, also symplectic. The best general choice when accuracy matters more than matching the shipped scenarios exactly.

RK4
Fourth order and not symplectic: locally the most accurate of the three and the only one whose energy error grows steadily over many orbits.

Each scheme's convergence order is *measured* by the validation suite, not asserted, and the tolerances of the integrated checks are derived from it.

5.2 Dark matter

An optional halo adds a force term with an adjustable flat rotation speed. It changes the force law rather than the picture: switch it on with the rotation curve open and a falling curve flattens. It stays on until you switch it off, so a scenario loaded afterwards is running with modified gravity — the rotation curve panel says so.

5.3 Tides

Tidal stress is computed as the difference between the pull on a body's near side and the pull on its centre. Where that difference exceeds the body's own gravity, the body comes apart — which is what happens in the Tidal Disruption Event scenario, and what the Roche limit marks.

6 Scenarios

59 scenarios ship with Gravitas. Open the gallery from Scenario in the rail; each is tagged by curriculum concept, so the gallery can be scanned for the week you are teaching rather than read end to end.

Loading a scenario rebuilds the world from a seed, so the same scenario opens the same way every time, and a share link carries the seed with it.

6.1 Binary Systems

Binary Pair: two stars, one balance point

Two stars of two solar masses each, four AU apart, circling their common center of mass once every four years. Neither one is stationary and neither one is orbiting the other: both go round the same point in between them. Watch the trails and the balance point gives itself away.

Binary Planet Lab: a planet around one star of a pair

Two stars of one and half a solar mass, ten AU apart on an orbit of eccentricity 0.4, and an Earth-mass planet circling the heavier one at 1.5 AU. Nothing here is randomized: every mass, distance and starting angle is written down, so two runs differ only where you make them differ. Move the planet outward and there is a distance past which it stops coming back. Finding roughly where, and how much of that answer is physics rather than timestep, is the experiment.

Circumbinary Planet Lab: a planet around both stars at once

The same two stars, with the planet outside them both at 40 AU, orbiting the pair as though it were one object. Real planets do this - Kepler-16b is the famous one - but only far enough out that the two stars begin to look like one. Bring it inward and the approximation fails: the changing pull of a binary it can still tell apart walks its orbit outward over a few dozen binary periods until it leaves. Here the boundary is a floor, not a ceiling.

Binary Stars

A pair of suns in mutual orbit with 5 planets orbiting the binary system. The complex gravitational environment creates interesting orbital dynamics and potential habitable zones. Similar to real binary star systems like Alpha Centauri.

6.2 Chaos & Encounters

Three-Body Sensitivity Lab: the same start, twice

Three six-solar-mass stars at the corners of an equilateral triangle, rotating rigidly about their common centre. This is an exact solution of the three-body problem, found by Lagrange in 1772, and for three equal masses it is unstable: the triangle holds for a few turns and then comes apart. Built for one experiment - run it twice from starts that differ by fifteen hundred kilometres in a system a hundred and thirty million kilometres across, and watch how long the two runs stay together.

Gravity Slingshot

A massive black hole (60 $M_\odot$) paired with a smaller companion (3 $M_\odot$) create dramatic gravitational assists for nearby planets and gas giants. Watch objects gain tremendous velocity through close encounters, mimicking spacecraft gravity assists.

Rogue Encounter

A wandering black hole (30 $M_\odot$) passes through a stable planetary system with 12 planets, 4 gas giants, and asteroids. Watch the dramatic orbital disruption, planet ejection, and tidal capture events as the rogue intruder wreaks havoc.

Star Frisbee

A dense stellar disk thrown past a rogue black hole. Will it be shredded or survive the flyby?

Kessler Cascade

Hundreds of micro-stars orbiting chaotically, colliding and ejecting like a debris cloud.

Alien Dyson Swarm Collapse

A hypothetical Dyson swarm of artificial satellites falls into a black hole after a catastrophic orbital failure.

Slingshot Gauntlet

A fast-moving star fired through a black hole obstacle course. Watch gravitational slingshots.

6.3 Compact Objects

Black Hole Lab: a ten solar mass hole, and four things orbiting it
A single stellar-mass black hole with four bodies on stable circular orbits around it. Nothing is falling in. Gravity a long way from a black hole is the same gravity as anywhere else, and an object with sideways motion goes round it exactly as it would go round a star of the same mass. Select the black hole to read its Schwarzschild radius, its average density on that scale, its Hawking temperature and how long it has left.

GW150914: First Gravitational Wave Merger
Simulates the historic merger of two massive black holes (36 & 29 $M_\odot$) detected by LIGO in 2015. Watch as they spiral together, emit gravitational waves, and merge into a single, more massive black hole.

Binary Black Hole
Two stellar-mass black holes (15 & 10 $M_\odot$) locked in mutual orbit with spectacular relativistic jets. Watch as they spiral together, create gravitational waves, and eventually merge into a single, more massive black hole. The jets point in random directions for each black hole, creating a dynamic cosmic display.

Triple Black Hole
A chaotic three-body dance of massive black holes (20, 15, & 10 $M_\odot$) in a complex orbital arrangement. This unstable configuration will eventually eject one black hole while the remaining two merge. Demonstrates the chaotic nature of multi-body gravitational systems.

Supermassive Core
One enormous black hole (80 $M_\odot$) dominates a dense stellar swarm with 50 planets, 5 gas giants, and 100 asteroids. The intense gravitational field creates spectacular accretion disks and tidal disruption events. Similar to the environment around real supermassive black holes in galactic centers.

Sagittarius A*
The Milky Way's central supermassive black hole (4000 $M_\odot$, scaled down for simulation) with fast-moving S-stars, compact objects, and debris in extreme orbits. Witness the incredible gravitational forces and relativistic effects near our galaxy's supermassive black hole.

Neutron Star Merger
Two neutron stars (1.4 $M_\odot$ each) spiral toward each other in a death dance. This rare event produces gravitational waves, gamma-ray bursts, and creates heavy elements through r-process nucleosynthesis. Based on the LIGO-detected GW170817 event.

Pulsar with Planets
A rapidly spinning neutron star with 3 planets in tight orbits. The pulsar's intense magnetic field and radiation create a harsh environment. Based on the first confirmed exoplanets discovered around PSR B1257+12.

White Dwarf Binary
Two white dwarf stars in a close binary system with accretion between them. One star gradually steals material from its companion, potentially leading to a Type Ia supernova. Includes debris disk and stellar remnants.

Stellar Graveyard
A dynamic collection of stellar remnants: 3 black holes, 5 neutron stars, and 8 white dwarfs with surviving planets and extensive debris fields. Watch these stellar corpses interact in their final gravitational dance.

Compact Object Zoo
A diverse collection of compact objects: multiple black holes, neutron stars, and white dwarfs of various masses interacting in a dense environment. Perfect for studying the different types of stellar endpoints and their interactions.

Millisecond Pulsar
An extremely fast-spinning neutron star (recycled pulsar) with a white dwarf companion and planetary debris. These 'recycled' pulsars are spun up by accretion and are among the most precise timekeepers in the universe.

Intermediate Mass BH
A rare intermediate-mass black hole (400 $M_\odot$) in a globular cluster environment with dense stellar populations. These elusive objects bridge the gap between stellar-mass and supermassive black holes.

Micro BH Swarm
A dynamic swarm of small black holes (0.6-1.8 $M_\odot$) with planets and gas giants in chaotic orbital dance. Watch as these stellar-mass black holes interact, merge, and create complex gravitational resonances.

Hungry Hungry Holes
Four black holes at the corners of a square, pulling stars from a shared central cluster.

Black Hole Billiards
A few small black holes orbiting a supermassive one, perturbing each other and creating chaotic motion.

6.4 Exoplanets

TRAPPIST-1 System
A compact planetary system with seven Earth-sized worlds orbiting a cool red dwarf star just 40 light-years away. All planets are packed close to their tiny sun, with several in the habitable zone. Can you keep this delicate system stable?

Transit Lab: HD 209458 b
The first exoplanet ever caught crossing its star, found in 1999 after radial velocities said where to look. A hot Jupiter on a 3.5-day orbit, drawn here at true relative scale: the star is 1.155 solar radii, the planet 1.38 Jupiter radii, and the silhouette on screen is the same 12% radius ratio the light curve reports. Open the Light Curve panel and watch the 1.7% dip repeat.

Exoplanet Characterization Lab
HD 209458 again, but with the star free to move. In the Transit Lab it is pinned so the light curve stays centred; here both bodies orbit their common centre of mass, which is what the radial-velocity and astrometry instruments need in order to measure anything. The star circles a point 2.7 millionths of an AU away at 84 metres per second: far too small to see and easily large enough to detect. Open Radial Velocity and watch the wobble that found this planet.

Blended Binary: a hidden companion
The same star and planet as the Transit Lab, with a second star half a magnitude fainter sitting 300 AU away: far too close on the sky for a survey telescope to separate, and well inside one photometric aperture. Its light fills in part of the dip, so the transit measures shallower and the planet looks smaller than it is. Correcting for exactly this effect is what high-resolution imaging surveys of planet hosts are for.

Exoplanet Lab
A diverse collection of 120+ exoplanets, gas giants, and even pulsar planets around various stellar hosts. Explore the incredible diversity of planetary systems with interactive orbital mechanics and planetary interactions.

6.5 Galaxies & Clusters

Spiral Galaxy: what we expected
A galactic bulge with ninety stars orbiting it, each launched at exactly the speed the visible mass says it should have. This is the prediction, not the observation: with the mass concentrated in

the middle, orbital speed falls away as the inverse square root of radius, the same way it does across the Solar System. Open the Rotation Curve panel and read the slope. Then load Milky Way Rotation and read it again.

Milky Way Rotation: what we actually see
The same disc, with every star moving at the same speed no matter how far out it is. That is what telescopes measure in real spiral galaxies, and it is far too fast for the stars you can see to hold on to: this scenario starts with a dark-matter halo switched on, because without one the disc does not survive. Open the Rotation Curve panel and switch the halo off to watch it come apart.

Coma Cluster: Zwicky, 1933
Twenty-four galaxies swarming in a bound cluster, on randomly oriented orbits, named after the members of the real Coma Cluster that Fritz Zwicky measured. He added up the light, added up the motions, and found the second answer hundreds of times larger than the first. He called the difference dunkle Materie and was ignored for forty years. Select a galaxy to read its speed, and work the same calculation he did.

Dense Star Cluster
A gravitationally bound collection of main-sequence stars, evolved giants, and stellar remnants with mutual gravitational interactions. Watch stellar encounters, binary formation, and the dynamic evolution of this stellar community over time.

Galactic Center
A supermassive black hole (4000 $M_\odot$) surrounded by high-velocity stars, stellar remnants, and dense stellar populations. Experience the extreme gravitational environment with spectacular accretion, jets, and relativistic effects.

Galactic Collision
Two supermassive black holes (1.2M & 1.0M $M_\odot$) with hundreds of stars representing galactic cores in collision. Witness the formation of tidal streams, stellar disruption, and the eventual merger of supermassive black holes.

Quasar Cannon
A supermassive black hole is actively feeding on a dense star cluster. Watch a beam of light form as stars spiral inward.

The Pinwheel Galaxy Core
Two intermediate black holes in the center of a stellar disk. The disk forms a rotating pinwheel pattern as stars are slung around.

Tidal Arm Tango
Two black holes dance past each other, flinging stars into massive tidal arms like colliding galaxies.

6.6 Habitability

Habitable Zone Lab: the inner Solar System, with the zone drawn
The Sun with Venus, Earth, Mars and Ceres on their real orbits, and the circumstellar habitable zone drawn around the star from a published prescription. Venus sits inside the inner edge and Mars outside the outer one on the conservative definition, and only one of the four has liquid water on its surface today. Switch the Habitable Zone Model setting to see the optimistic band, which reaches out past Mars.

6.7 Orbits & Kepler

Gravity Assist Lab: one planet, one spacecraft, nothing else
A rogue planet of five Jupiter masses drifting through empty space, and an Earth-mass spacecraft crossing its path. No star, which is the point: with nothing else in the universe the planet

moves in a straight line and its frame is exactly inertial, so the strangest fact about a gravity assist can be stated exactly rather than approximately. The spacecraft leaves at the same speed relative to the planet that it arrived with, and at a completely different speed relative to everything else.

Gravity Assist: the same flyby, with a Sun to steal from
The same planet, now on a circular orbit five AU from a star, and a spacecraft that meets it. This is where the energy comes from: the spacecraft leaves faster around the star than it arrived, and the planet is slowed by exactly the momentum it gained. Everything here is approximate - the planet is accelerating, so its frame is not inertial and the encounter is only locally two-body - and the panel reports how approximate rather than hiding it.

Lagrange Point Lab: two stars, one circular orbit, one test particle
A sunlike star and a companion a thirtieth its mass, on an exactly circular orbit eight astronomical units across, with a tracer light enough that neither notices it. This is the circular restricted three-body problem: the simplest system with Lagrange points, forbidden regions and a conserved Jacobi constant, and the only arrangement in which any of those are exactly true.

Orbital Transfer Lab: one star, one spacecraft, one destination
A spacecraft on a circular orbit at 1 AU and a station on a circular orbit at 2.5 AU, about a single sunlike star with nothing else in the system. Circular and coplanar is what makes the transfer between them exactly solvable, so a burn planned by hand can be checked against the arithmetic rather than only watched.

Kepler's 2nd Law - Equal Areas
A planet in a nearly circular orbit and an eccentric orbiter around a central star. The area sweep visualization starts automatically for the eccentric body - watch how the wedges change shape but maintain equal area, showing why objects move faster at periapsis than at apoapsis.

6.8 Orbital Resonance

Galilean Resonance: the Laplace lock
Io, Europa and Ganymede on the 4:2:1 chain Laplace explained in 1805, with Callisto outside it for contrast. The periods are not exactly 4:2:1 — Europa takes 0.37% longer than two Io years — and that near miss is the point: what holds them is the Laplace argument, which stays near 180 degrees instead of running through every value, so the three are never all in conjunction. Callisto sits within 0.03% of 7:3 with Ganymede and is in no resonance at all. A scale model, 100 times life size.

Broken Laplace Resonance: one percent out
The same four moons with one number changed: Europa starts one percent further from Jupiter. That is a hundred times wider than the resonance can hold, and the lock fails — the Laplace argument stops swinging about 180 degrees and starts going all the way round, once every forty-six Io orbits. Run it beside Galilean Resonance and the difference between a system that is locked and one that merely has convenient periods is on screen in under a minute.

Pluto and Neptune: the 3:2 that protects
Pluto's orbit crosses Neptune's, and they have never come close. The 3:2 resonance is why: Pluto goes round twice for every three Neptune years, and the resonant argument librates about 180 degrees rather than circulating — so every conjunction happens near Pluto's aphelion, far outside Neptune's reach. A third body runs almost the same orbit from outside the resonance; watch what becomes of it. True scale; one second is about 270 years.

Jupiter Trojans: the 1:1 co-orbital resonance
Two of the five Lagrange points are places a body can sit still relative to Jupiter forever, and thousands of asteroids do. A probe placed exactly at L4 does not move in Jupiter's rotating

frame; the Trojan 617 Patroclus loops around L5 once every twelve and a half Jupiter years. A third probe starts one degree from L3 — an equilibrium too, and unstable — and leaves. The fourth runs a wider ordinary orbit and is in no resonance at all.

6.9 Solar System

Solar System
A simulation of our Solar System featuring real planets with correct masses, orbital distances, diameters, and colors. Includes Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune with their actual properties, plus real asteroids (Ceres, Vesta, Pallas) and famous comets (Halley, Hale-Bopp, Hyakutake) with authentic orbital periods and characteristics.

Retrograde Mars: the loop that needed epicycles
The Sun, Earth and Mars at their real distances and periods, and nothing else. Watched from outside, both planets go round the Sun the same way and never turn back. Switch the reference frame to Earth, in Tools, and Mars stops circling and starts drawing a loop that doubles back on itself. Nothing about the physics changed; only the frame did. That loop is the observation Ptolemy built epicycles to reproduce and Copernicus explained away, and here you can turn it on and off with one control.

Earth-Moon System
A detailed simulation of the Earth-Moon system with accurate masses, orbital mechanics, and realistic appearances. Features Earth with its blue oceans and green continents, and the Moon with its characteristic gray surface and craters. Perfect for studying orbital dynamics and tidal effects.

Interstellar Visitor: 1I/'Oumuamua
The first object ever seen passing through the Solar System from somewhere else, on its measured orbit: perihelion inside Mercury, eccentricity 1.20, and 87.7 km/s at closest approach. Earth is shown for scale. Select the visitor and look at the sign of its total energy: it is positive, which is the whole story. It is not bound to the Sun, it was never going to stay, and it will not be back.

Kuiper Belt
An accurate simulation of our Solar System's Kuiper Belt featuring real dwarf planets (Pluto, Eris, Haumea, Makemake), large KBOs (Quaoar, Sedna, Orcus, Varuna), and smaller objects (Ixion, Huya, 2002 AW197) with realistic masses and orbital properties.

6.10 Stellar Evolution

Supernova Remnant
The explosive aftermath of a massive star death: a neutron star surrounded by high-velocity debris, shocked planets, and disrupted gas giants. Experience the violent and energetic environment left behind by stellar death.

Stellar Nursery
A dense cluster of young stars around a proto-black hole. Watch interactions and ejections as the cluster evolves.

6.11 Tides & Disruption

Tidal Disruption
Multiple objects approach a supermassive black hole (2000 $M_\odot$) and are torn apart by extreme tidal forces. Watch as planets and gas giants are stretched, disrupted, and either ejected or accreted, creating spectacular debris streams.

7 Guided investigations

22 investigations, 618 steps between them, 345 of those graded, against 137 stated learning objectives. Open them from Learn in the rail.

Every investigation follows the same discipline:

1. It asks for a **prediction** before it shows anything.
2. It hands over an **instrument** and asks for a measurement.
3. It **plots the student's own readings** back to them.
4. It **saves progress** in the browser, so a session can be left and resumed.
5. It **exports a lab report** as a PDF, which the student submits through whatever system the course already uses.

Nothing is submitted anywhere by Gravitas itself. There is no server to submit to.

Investigation	Steps	Graded	Time
Kepler's Laws	23	13	35-45 min
Why Mars Goes Backwards	33	18	35-45 min
Finding Planets by Their Shadows	29	13	50-70 min
Bound, Unbound and Escape	23	9	35-45 min
Weighing the Stars	35	17	35-45 min
Black Holes by the Numbers	29	17	35-45 min
Finding Planets by Their Tug	37	20	45-55 min
The Goldilocks Question	37	19	40-50 min
The Missing Mass	33	17	45-60 min
Tides	30	16	35-45 min
The Butterfly Effect in Space	28	13	55-70 min
When Orbits Lock	33	16	55-70 min
Can You Detect This Planet?	26	13	30-35 min
Design the Schedule	14	7	35-40 min
Planets in Binary Stars	37	21	40-50 min
Where Does a Gravity Assist Get Its Speed?	22	13	15-20 min
Getting There From Here	20	12	20-25 min
Where Can It Get To?	19	8	25-30 min
What Is a Gravitational Wave?	24	14	30-40 min
Listening to Spacetime	24	15	60-75 min
A Universe of Stars	28	25	70-90 min
Lives of Stars	34	29	80-100 min

7.1 Kepler's Laws

Measure the shape, pacing and timing of real orbits 23 steps, 13 of them graded, 35-45 min, Introductory astronomy.

Work through all three of Kepler's laws by measuring orbits rather than being shown them: find the focus of an ellipse, watch equal areas sweep out in equal times, and recover the three-halves power law by plotting it yourself.

7.2 Why Mars Goes Backwards

Change the frame, and fourteen centuries of epicycles fall away 33 steps, 18 of them graded, 35-45 min, Introductory astronomy.

Twice every three years Mars stops in the sky, reverses, and loops back on itself. Watched from outside, nothing of the sort happens: Earth and Mars both go round the Sun the same way and never turn back. You will measure both orbits, predict what Mars does when seen from Earth, then switch the reference frame and watch the loop draw itself. Nothing in the physics changes when you do. That is the entire point, and it is what took astronomy from Ptolemy to Copernicus.

7.3 Finding Planets by Their Shadows

Measure a transit, weigh what it tells you, and find what is hiding 29 steps, 13 of them graded, 50-70 min, Introductory astronomy.

Work through the transit method from first principles on HD 209458 b, the first planet ever caught crossing its star: measure a depth and turn it into a radius, correct it for limb darkening, time two transits to get a period, read an atmosphere out of the color of the dip, and finish by finding the hidden companion star that makes the planet look smaller than it is.

7.4 Bound, Unbound and Escape

Find out what decides whether something comes back 23 steps, 9 of them graded, 35-45 min, Introductory astronomy.

Fire something off a planet and find out what decides whether it falls back, circles forever, or leaves and never returns. Work up from the experiment to the idea behind it: every object near a star carries an amount of energy, and the sign of that one number settles the question. Finish on a real interstellar visitor and decide for yourself whether it will be back.

7.5 Weighing the Stars

Use an orbit to measure something you cannot put on a scale 35 steps, 17 of them graded, 35-45 min, Introductory astronomy.

Kepler's laws end with Newton's correction, and this is what that correction is for. Watch two stars circle each other, find the balance point they are both going round, and use nothing but the size and the timing of their orbit to work out how much each one weighs. No telescope has ever put a star on a scale; this is how it is actually done.

7.6 Black Holes by the Numbers

Make a black hole bigger and discover some surprising rules 29 steps, 17 of them graded, 35-45 min, Introductory astronomy.

Change one thing about a black hole, its mass, and watch four completely different properties respond. Its event horizon grows in step with the mass. Its average density falls. It gets colder. It lives dramatically longer. Two of those four surprise almost everybody, and you will predict them before you measure them.

7.7 Finding Planets by Their Tug

Watch a star wobble, weigh its planet, and combine the clues 37 steps, 20 of them graded, 45-55 min, Introductory astronomy.

A planet you cannot see still pulls on its star, and the star moves. Measure that motion two different ways, turn it into a mass, and combine it with the radius a transit gave you to work out what kind of world it is.

7.8 The Goldilocks Question

Move a planet, change its star, and decide what 'habitable' really means 37 steps, 19 of them graded, 40-50 min, Introductory astronomy.

Work out for yourself why a planet twice as far from its star receives a quarter as much energy, why dim stars have their habitable zones tucked in close, and why an eccentric orbit means a planet does not receive one steady amount of light all year. Then finish with the harder question the phrase "habitable zone" invites people to skip: what does being inside it actually tell you?

7.9 The Missing Mass

Weigh a system twice, and find the two answers do not agree 33 steps, 17 of them graded, 45-60 min, Introductory astronomy.

There are two ways to weigh a system in space: add up the light, or watch how things move. For the Solar System the two agree. For a galaxy they do not, and for a cluster of galaxies they are out by more than a factor of ten. Students arrange mass and watch the rotation curve it makes, turn a measured speed into an enclosed mass, then take a real galaxy's curve and try to fit it with stars alone — and fail, in the specific way the field failed for a decade, before adding a halo and getting it right. It closes on Zwicky's cluster and the mass budget of the universe. It is how dark matter was found, and it is a measurement rather than a theory.

7.10 Tides

Stretch a world, move a moon, and discover why gravity can tear objects apart 30 steps, 16 of them graded, 35-45 min, Introductory astronomy.

Tides are not caused by strong gravity. They are caused by gravity being unequal across an object, and the whole lesson is built on that one subtraction: take the pull on the centre away from the pull on the near side and the far side, and everything from the two daily high tides to a star being shredded by a black hole falls out of what is left.

7.11 The Butterfly Effect in Space

Run the same system twice and find out how long the answer lasts 28 steps, 13 of them graded, 55-70 min, Introductory astronomy.

Two runs of the same three stars, started from positions differing by fifteen hundred kilometres in a system a hundred and thirty million kilometres across, end up somewhere completely different. Nothing random happens in between: the simulation is deterministic, and running it twice from exactly the same numbers gives exactly the same answer both times. Along the way you will measure a case that looks like chaos and is not, put a number on how fast prediction fails, and check that the number is a property of the physics rather than of the computer.

7.12 When Orbits Lock

A ratio is a hint. Find out what counts as proof 33 steps, 16 of them graded, 55-70 min, Introductory astronomy.

Three of Jupiter's moons keep time with each other, Pluto crosses Neptune's orbit and has never come near it, and thousands of asteroids sit sixty degrees ahead of Jupiter and stay there. All three are the same phenomenon, and none of them is explained by the thing everybody quotes: the ratio of the periods. You will measure the ratios, find that the tidiest one in the system belongs to a moon in no resonance at all, and then measure the quantity that actually settles it — an angle that either swings or goes round.

7.13 Can You Detect This Planet?

Two methods, the same problem: the answer was decided before the data arrived 26 steps, 13 of them graded, 30-35 min, Introductory astronomy.

A planet is either there or it is not, but whether you find it depends on choices you make before you take a single measurement. Plan two radial-velocity runs of the same star with the same instrument and the same number of nights, and find that one detects a Jupiter and the other cannot tell you anything. Then do it again with transits, where the same planet is a 587-sigma certainty from space and a 4-sigma maybe from three nights on the ground — and work out how many more nights would fix that, and where the answer stops improving.

7.14 Design the Schedule

Eight nights on the real instrument, and the times you choose decide the answer 14 steps, 7 of them graded, 35-40 min, Introductory astronomy.

You have eight nights and one star. Plan the run yourself in the live Radial Velocity panel, commit to a prediction, then observe two schedules side by side against the same star with the same instrument and the same noise — and watch one of them recover a Jupiter while the other cannot establish that the velocity changes at all. Then break your own result: change the seed, lose a fortnight to weather, and type a list of dates by hand, until you can say what a reported period has to carry before anybody else can check it.

7.15 Planets in Binary Stars

What survives around two stars, and how you would know 37 steps, 21 of them graded, 40-50 min, Introductory astronomy.

Most stars come in pairs, so most planets have to make a living in a system with two suns. Some orbits work and some do not, and the line between them is sharper than you would guess. Find it twice — once for a planet around one star, once for a planet around both — and then find out how much of what you just measured was the physics and how much was the arithmetic.

7.16 Where Does a Gravity Assist Get Its Speed?

The same flyby, measured in two frames, with two different answers 22 steps, 13 of them graded, 15-20 min, Introductory astronomy.

Voyager 2 arrived at Jupiter travelling ten kilometres a second and left travelling twenty-six. Jupiter did not burn any fuel for it. Fly the same manoeuvre yourself, measure it in the planet's frame and in an inertial one, run it past both sides of the planet at once, and find out why the two measurements disagree — and who actually paid.

7.17 Getting There From Here

Two burns, a long coast, and the arithmetic that decides both 20 steps, 12 of them graded, 20-25 min, Introductory astronomy.

A spacecraft at 1 AU, a station at 2.5 AU, and no fuel to waste. Work out both burns and the coast between them with a pencil, then fly the manoeuvre and see whether the engine agrees with you. It does — to a part in a thousand — which is what makes the two surprises in it worth trusting: you speed up to go further out, and you have to speed up again on arrival or you fall straight back.

7.18 Where Can It Get To?

Forbidden regions, five balance points, and one conserved number 19 steps, 8 of them graded, 25-30 min, Introductory astronomy.

Two stars on a circular orbit and a speck of dust that feels them both. There is one number you can compute about the speck that tells you where it is forbidden to be — and as you make it go faster, walls open one at a time in a fixed order. Find the five places where the speck could sit still, work out which of them it can reach, and then find out why "can reach" is three different questions wearing the same coat.

7.19 What Is a Gravitational Wave?

A first look at what moves, what travels and what a detector feels 24 steps, 14 of them graded, 30-40 min, Beginner, no physics background needed.

Two objects circle each other on screen and emit no light at all. Over twenty-four short screens you work out what leaves them, what it does to anything it passes, and how an instrument could notice - and you learn to tell the three kinds of picture apart: the drawing, the calculation and the measurement. No equations, no prior physics, and it can be done with the sound off.

7.20 Listening to Spacetime

Work out what made a signal, then check it against the real thing 24 steps, 15 of them graded, 60-75 min, Introductory astronomy.

A pattern arrives with no label on it: a wiggle that gets faster and louder and then stops. Over twenty-four steps you work out what could produce it, measure the two relationships that give it away, find out which questions the model can answer and which it cannot, and finish by comparing your answer with what two detectors in Louisiana and Washington actually recorded in September 2015. You can do all of it with the sound off.

7.21 A Universe of Stars

Size, colour and the H-R diagram, from eight modelled stars 28 steps, 25 of them graded, 70-90 min, Introductory astronomy.

Three stars, no labels, and a guess about which is biggest. Over twenty-eight steps you separate the four things that get confused with each other - mass, radius, temperature and luminosity - learn to read the diagram that organises them, meet giants and supergiants and white dwarfs where they actually sit on it, work out why the heaviest stars live the shortest lives, and finish by counting a synthetic population twice to see why the stars you can see are not the stars there are.

7.22 Lives of Stars

From clouds to cosmic remnants, along eight published tracks 34 steps, 29 of them graded, 80-100 min, Introductory astronomy.

A star is not a thing so much as a process that takes a while. Over thirty-four steps you follow three of them from a contracting cloud to what they leave behind — a solar-mass star to a white dwarf, a ten solar-mass star to a neutron star, and a forty solar-mass star to a black hole — reading every stage off the same diagram and the same published tracks. You will also meet the star that does none of this: a red dwarf that will still be burning hydrogen when the Universe is a hundred times its present age.

8 Running a controlled experiment

The A/B Bench, in Tools, is for the question a sandbox cannot otherwise answer: *what does changing this one thing actually do?*

Watching a simulation twice does not answer it, because the second run never starts quite where the first did. The bench makes it start there exactly.

1. **Capture start.** The world as it stands becomes the experiment's origin — its seed, its clock, its bodies and their identities, the integrator, the reference frame and the observer.
2. **Name it, and choose what to measure.** Bodies as chips, quantities as checkboxes. A quantity needing two bodies stays greyed out until two are chosen.
3. **Record Run A.**
4. **Return to start**, exactly. The simulation holds there.
5. **Change one thing.**
6. **Record Run B.**
7. **Compare.** The two runs are drawn on one simulated-time axis, with a table of absolute and fractional differences.

What can be measured: position and trajectory, separation between two bodies, speed and velocity, distance from a chosen primary, orbital period, closest approach, total energy, angular momentum, and the numerical drift in the last two.

8.1 What the bench will tell you off for

The bench names the parameter that differed between the runs, and it says so when more than one did — a comparison with two independent variables cannot say which one caused the difference. You can confirm that you meant it, because a deliberate multivariable comparison is a legitimate thing to run.

It does not count elapsed time, the camera, the theme or a trail length as variables. It *does* count the simulation speed, because the speed control scales the integration step and therefore changes the numerical error a run reports, and it counts the seed and the scenario, because either one makes it a different world.

8.2 Honest sampling

Two runs are never sampled at the same instants: an animation frame takes as long as it takes, and a scenario that substeps advances its clock several times inside one frame. So the bench records each sample with its own simulated time and aligns the two runs on that, interpolating between samples and refusing to extrapolate past the end of either. Where the runs do not overlap in simulated time it says there is no comparison, rather than drawing one.

The recording indicator shows both the number of samples and the simulated seconds they cover, because those two are not interchangeable.

8.3 Keeping and sharing an experiment

Experiments can be renamed, duplicated, deleted, saved in the browser and reopened. Saved experiments are bounded — if one is too large, or the store is full, the bench says which and points you at the export, rather than failing quietly.

Exporting gives two files: a **CSV** of both runs, one row per sample, with the run named in a column and the unit in every column heading; and a **JSON manifest** carrying the experiment's definition and its provenance — scenario, seed, integrator, timestep, units, a hash of the initial state, what was measured and what changed. The manifest is the file to paste into a lab report, and reopening it restores the experiment's setup so somebody else can run it themselves.

Share setup puts the same thing in a link. The link carries the starting state and the A/B difference; it never carries the recorded runs, so what arrives is an experiment to perform rather than somebody's results.

9 Capturing, exporting and sharing

9.1 Screenshots and clips

Capture → Screenshot (or `P`) saves a PNG with the scenario's name, the scale bar and the elapsed simulated time drawn into the image itself, so a figure in a lab report documents its own scale and its own clock without a caption having to supply them.

Record Clip records a stretch of the run to a video file: H.264 in MP4 where the browser can encode it, WebM where it cannot. The same three facts are burned into every frame. A recording indicator shows the elapsed time and the size, and a take stops itself at three minutes or 80 MB, whichever comes first.

9.2 Data

Export CSV (or `E`) writes the recorded timeline: positions, velocities and energies over time, plus the light curve if it is running. Every column names its own unit — `t_days`, `x_au`, `vx_kms`, `r_au`, `E_kin_J` — and each row records what the body is orbiting, so a period can be fitted without first having to work out what the numbers mean. A companion notebook that reads the file ships with the project.

9.3 Share links

Any configuration encodes into the URL. Hand out `gravitas-sim.online/#…` as an assignment and every student opens the identical system, down to the seed; a student can send one back as an answer. Nothing touches a server.

9.4 Embed mode and lecture mode

Adding `?embed=1` to a link produces a chrome-free figure sized to its frame, which drops into a course page or a learning-management system in an iframe. It keeps its scale bar and its clock, and drops the panels that would not fit.

Lecture Mode fills the screen for projection: larger type and controls, the Daylight theme, a spotlight pointer, and arrow keys that step through a prepared sequence of links. `Esc` leaves.

10 Languages

The interface ships in 2 languages — English, Español — from a catalog of 3456 strings, and all 22 investigations are translated.

A translation carries words and nothing else. No scenario name, seed, widget identifier, numeric answer or measurement probe can be reached from a locale file, so a mistranslation can change what a student reads but never what the simulation measures or what counts as a correct answer.

11 For instructors

Instructor guides and answer keys for all 22 investigations, an adopter's guide and a curriculum map are published at https://gravitas-sim.online/instructors/. They are generated from the lessons themselves at build time, so an answer key cannot disagree with the lesson it answers.

They sit behind a passphrase, and the honest description of what that means is on the page itself: the site is static, with no server to check a credential against, so the materials are AES-GCM encrypted at build time and decrypted in the browser. That is real protection against casual discovery and against a student who finds the URL. It is not protection against a determined attacker who has the ciphertext. Instructors can request the passphrase through the contact link on that page.

11.1 Citing Gravitas

If you use Gravitas in teaching or research, please cite it. The repository carries a `CITATION.cff`, which GitHub renders as a ready-made citation, and a Zenodo record.

12 Limitations

Stated plainly, because a teaching tool that hides these is a teaching tool that will embarrass someone in front of a class:

- The engine is **two-dimensional**. Inclination exists for the observing panels as a viewing geometry, not as a third spatial dimension the bodies move in.
- It is **Newtonian**. There is no general relativity; the gravitational-wave inspiral is a fitted energy loss and the innermost stable circular orbit is drawn, not derived.
- **Collisions are perfectly inelastic.** Two bodies that touch become one, conserving mass and momentum but not shape, spin or debris except where a scenario models debris deliberately.
- **Long runs drift.** Any integrator accumulates error. The readout reports that drift live rather than hiding it, and the validation suite measures how fast each scheme accumulates it.
- **Some scenarios switch off part of the model on purpose** — static black holes, one-way gravity, the dark-matter halo. Those conserve less than the full model does, and the scenario audit names which ones and why.

13 Accessibility

Of the application:

- The controls are real focusable elements, and the actions you need during a run — pause, reset the view, screenshot, export, the shortcut list — have keys of their own.
- The simulation canvas carries a text description, and state that only the picture would otherwise show is announced through a live region.
- Where colour carries meaning it is not carrying it alone: the run-state indicator shows a coloured dot *and* the word, and every arrow in the vector overlay is named in the key beside it.
- Four themes, of which the default is near-black and high contrast and Daylight is its light counterpart. Windows high-contrast mode is honoured rather than overridden.
- Animation respects the operating system's reduced-motion setting.

Of this document: it declares its language, its text extracts and searches correctly, its structure is a real heading hierarchy with a bookmark outline, it contains no images and therefore nothing that needs alternative text, and its links are coloured dark enough to hold contrast against the page.

It is *not* a tagged PDF. The LaTeX distribution used to build it predates the tagging support that PDF/UA requires. A reader who needs tagged structure is better served by the web documentation, which is HTML.

14 Where to find more

The application
https://gravitas-sim.online

What the model does and does not claim
https://gravitas-sim.online/model/

Instructor materials
https://gravitas-sim.online/instructors/

Source, issues and the technical documentation
https://github.com/gravitas-sim/gravitas-sim.github.io

This manual was generated from the application's own scenario catalog and investigation manifest. If a count in it disagrees with what you see on screen, that is a bug: please report it.